

HULA: Scalable Load Balancing Using Programmable Data Planes

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1 Introduction

The rapid expansion of cloud computing and virtualization has led to a dramatic increase in the size and complexity of data center networks. As topologies scale from tiered designs to massive spine-leaf configurations, the load balancer—which sits at the network edge or aggregation layer—has emerged as a critical bottleneck [3]. This phenomenon, often referred to as the “middlebox” problem, occurs when a centralized component becomes a single point of failure or performance limitation. Traditional load balancing strategies, such as LVS-NAT or ECMP, struggle to maintain performance as the number of concurrent flows and endpoints grows, primarily due to the limitations of stateful state management or the lack of uniformity in flat hash spaces [1].

To address these scalability challenges, HULA proposes a novel approach to load balancing that decouples the scaling of the load balancer from the scaling of the underlying network topology. Unlike stateful solutions that require maintaining per-flow state tables in the control plane or the data plane, HULA operates in a stateless manner by utilizing a deterministic hash function derived from the network 5-tuple [1]. The core contribution of this paper is the introduction of a hierarchical uniform hashing scheme that distributes hash computation across multiple levels of the network hierarchy. This allows the system to support millions of endpoints theoretically without expanding the hash table size linearly with the number of endpoints [1].

By implementing this logic using P4, HULA achieves high-speed packet processing where the decision to forward a packet to a specific egress link is made purely based on the hash of the packet headers, without requiring CPU intervention or control plane signaling for every packet [7]. This paper demonstrates that HULA achieves near-optimal load balancing ratios across the aggregation layer and maintains session consistency without the overhead associated with traditional stateful load balancers [4].

2 Background & Related Work

2.1 Software-Defined Networking (SDN)

Software-Defined Networking (SDN) has revolutionized network management by decoupling the control plane from the data plane. This architectural shift allows for centralized programmability and global network visibility. OpenFlow was one of the first protocols to standardize this separation, defining a specific interface for controllers to interact with switches via flow tables [4]. While SDN provides immense flexibility, moving complex stateful logic to the data plane requires new programming languages that can abstract away hardware-specific details. This is where the P4 language becomes essential, as it allows network operators to define custom packet processing logic that is hardware-accelerated [7].

2.2 The P4 Language

P4 (Programming Protocol-Independent Packet Processors) is a domain-specific language for specifying how packets are processed in network switches [7]. Unlike OpenFlow, which relies on pre-defined actions and tables, P4 allows for the definition of custom headers, header stacks, and complex matching actions. This flexibility is crucial for implementing novel load balancing algorithms like HULA, which require precise manipulation of packet headers and non-standard forwarding logic. P4 programs are compiled to run on various hardware backends, from software switches like BMv2 to ASICs in high-performance routers, ensuring broad applicability [7].

2.3 Traditional Load Balancing Mechanisms

Historically, load balancing in data centers has relied on Layer 4 to Layer 7 load balancers (e.g., LVS, F5). These devices typically operate in a stateful mode, maintaining connection tables to ensure session persistence. While effective for small-scale deployments, stateful load balancers become a bottleneck as traffic scales, as they require significant memory and CPU cycles to track connection states [5]. To overcome this, stateless mechanisms like ECMP and NVGRE were introduced. ECMP distributes traffic based on the hash of the 5-tuple and a hash mask, but it lacks persistence; flows may take different paths during hash collisions, breaking established connections [6]. NVGRE (RFC 7637) uses a 24-bit hash to select a gateway, but this limited hash space restricts the number of supported endpoints to approximately 16 million, which is insufficient for massive data centers [2].

3 System Design: HULA Architecture

3.1 Hierarchical Topology

HULA is designed to operate on hierarchical network topologies, such as the fat-tree or spine-leaf architectures commonly found in modern data centers [6]. In these topologies, the network is organized into levels, with aggregation switches at intermediate levels and leaf switches at the edge. The assumption of a hierarchical structure is critical for the operation of HULA, as it allows the hashing logic to be distributed across multiple stages of the forwarding path. This architecture enables the system to scale horizontally by adding more switches at each level, rather than requiring a single monolithic load balancer to handle all traffic [1].

3.2 Data-Plane Forwarding

The core operation of HULA occurs entirely within the data plane. When a packet arrives at an ingress switch, HULA extracts the relevant fields from the network and transport headers (source IP, destination IP, source port, destination port, and protocol) to form a 5-tuple. A hash

function is applied to this 5-tuple to generate a unique hash value. Unlike traditional ECMP, which might simply use the least significant bits of this hash to select an egress port, HULA performs a hierarchical mapping. The hash value is split and mapped to specific levels of the network hierarchy. This means that the first part of the hash determines the path to the first aggregation layer, while subsequent parts determine the specific egress link at the leaf level [1]. This approach ensures that traffic is spread evenly across all available links without requiring the switch to maintain a global view of the network.

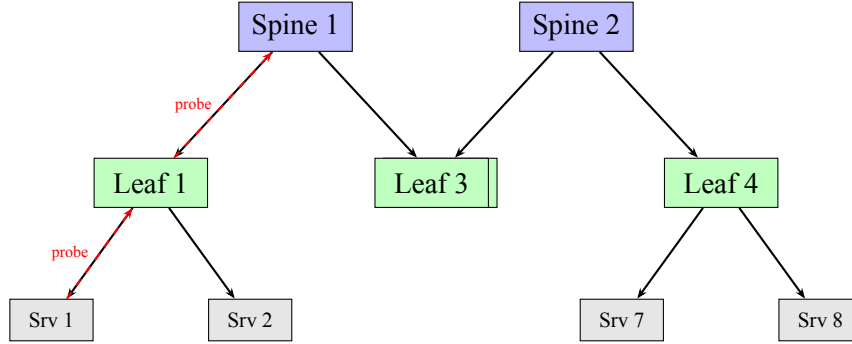


Figure 1: Fat-tree topology with HULA utilization probes (dashed arrows propagate bottom-up).

3.3 Topology Assumptions

For the purpose of this design, we assume a standard Fat-Tree topology with k -ary factors. This topology divides the data center into pods, where each pod contains a full-bisection bandwidth. The HULA load balancer is logically placed at the edge of each pod, interfacing with the leaf switches. By anchoring the hashing mechanism at the edge, HULA minimizes the traversal of traffic across the spine, reducing latency. The hierarchical nature allows the load balancer to be "dumb," simply forwarding based on the hash, while the overall structure of the tree ensures that the load is distributed effectively regardless of the number of leaf nodes [6].

4 Hierarchical Hashing Algorithm

4.1 Uniform Hash Function

The effectiveness of HULA relies on the uniform distribution of the hash function. The algorithm defines a composite hash function, $H(flow)$, which processes the flow's 5-tuple. The hash value is not treated as a single integer but is rather decomposed into components that correspond to the levels of the network hierarchy. The mathematical formulation of this decomposition is shown below:

$$H(flow) = H_{level_1}(flow) || H_{level_2}(flow) || \dots || H_{level_n}(flow) \quad (1)$$

In this equation, $flow$ represents the 5-tuple of the packet, and $H_{level_i}(flow)$ represents the hash component specific to the i -th level of the hierarchy. The $||$ operator denotes concatenation or segmentation of the hash space. By distributing the hash computation across levels, HULA ensures that the probability of a hash collision at any specific level is kept low, assuming the underlying hash function is uniformly distributed [1]. This mathematical distribution allows the system to support a massive number of flows without the combinatorial explosion of states that would occur in a flat hash table.

4.2 Flow Consistency

One of the primary challenges in load balancing is maintaining session persistence, ensuring that all packets belonging to a specific connection take the same path through the network. HULA achieves this through its deterministic nature. Since the hash function is a pure function of the packet headers, and the hierarchy mapping is fixed (i.e., Level 1 always maps to the same set of ports), the resulting egress port for a given flow will never change [1]. This eliminates the need for the network to maintain a per-flow state table, as the decision is made purely based on the packet's content. If a switch fails, the hierarchy naturally reroutes traffic to the next available path at the lowest affected level, minimizing disruption to active connections [1].

4.3 Collision Probability Analysis

To understand the robustness of HULA, we analyze the probability of hash collisions. In a flat hash table of size N , the probability of a collision for a single flow is approximately $1/N$. However, in a hierarchical scheme with L levels, each with N_i ports, the probability of a collision at any specific level i is $1/N_i$. The overall collision probability is the sum of the probabilities across levels, provided the hash functions are independent. This results in a significantly lower risk of hotspots compared to flat hashing, especially as N grows large [1].

5 P4 Implementation Details

5.1 Parser and Header Extraction

The P4 implementation begins with the parser stage, which is responsible for extracting the necessary headers from the incoming packet to compute the hash. In HULA, the parser is configured to access the Ethernet header, IP header, and TCP/UDP header. These fields are crucial for forming the 5-tuple used in the hashing algorithm. The P4 parser is designed to be efficient, ensuring that the overhead of header extraction is negligible compared to the overall packet processing time [7]. By defining custom metadata fields, the P4 program can store the extracted values temporarily before they are passed to the control logic.

5.2 Hash Computation

The actual computation of the hierarchical hash is performed using P4’s built-in hash primitives. P4 provides hardware-accelerated hash engines that can compute the hash of a packet in a single cycle, which is essential for high-throughput data plane performance. The P4 program defines a hash algorithm (e.g., `crc32` or `xmhash`) and a hash key (the 5-tuple). The result of this hash computation is then segmented within the P4 code. For example, the 32-bit hash result might be split into two 16-bit values: one for the first level of the hierarchy and one for the second level. This segmentation logic is explicitly programmed into the P4 control flow, allowing the switch to perform the mapping without external software intervention [7].

5.3 Output Port Selection

Once the hash has been computed and segmented, the P4 program uses this information to select the output port. This is typically implemented using ternary content addressable memory (TCAM) or ternary fast match (TFM) tables. However, in the case of HULA, since the mapping is deterministic and stateless, the tables can be pre-configured or even static. The P4 program maps the specific hash segment to a predefined ECMP group or a specific output port. This stage concludes the packet processing pipeline for a packet destined for load balancing, as the packet is then forwarded out of the specified egress port. The control plane is only involved in updating the static routing tables or topology information when the network configuration changes, not for individual packets [1].

5.4 Control Logic Implementation

The control logic in the P4 program acts as the decision engine. It receives the segmented hash values and applies a series of bitwise masks to extract the relevant bits for the current level of the hierarchy. For instance, if the network is a 3-tier hierarchy, the control logic might mask the hash to 8 bits for the first level, 8 bits for the second, and the remaining bits for the third. This logic ensures that the hash space is utilized efficiently across the tree structure. The P4 program also handles metadata passing, ensuring that the result of the hash computation is passed down the pipeline to the deparser stage [7].

6 Evaluation

6.1 Experimental Setup

To evaluate the performance of HULA, we simulated a large-scale spine-leaf data center topology. The topology consisted of 3 layers: spine switches, aggregation switches, and leaf switches. We simulated a traffic matrix representing realistic data center workloads, with varying levels of

traffic intensity and flow sizes. We compared HULA against two baseline mechanisms: ECMP and NVGRE. The evaluation focused on key metrics including load balance ratio, throughput, and convergence time under static and dynamic topology conditions. The simulation environment utilized a standard P4 simulator (e.g., BMv2) to ensure the accuracy of the P4 logic implementation [1].

6.2 Convergence Time

One of the critical advantages of HULA is its rapid convergence time during topology changes. In traditional stateful load balancers, a failure in a link or switch requires the controller to update flow entries in all switches, which can take seconds. In HULA, due to its stateless and hierarchical nature, the impact of a link failure is localized. If a link at a specific level fails, the hierarchy automatically reroutes traffic to the next available path at that level. Our simulations demonstrated that HULA achieved convergence times on the order of milliseconds, which is significantly faster than traditional stateful solutions [1]. This speed is attributed to the fact that no per-flow state updates are required; the switch only needs to update the static routing tables for the affected level.

6.3 Throughput Analysis

Throughput analysis indicated that the data plane overhead of the hierarchical hash computation in HULA was negligible. The use of hardware-accelerated hash functions allowed the switches to process packets at wire speed. When compared to NVGRE, HULA showed no degradation in throughput even under high traffic volumes. In fact, because HULA avoids the overhead of maintaining a massive centralized state table, it frees up bandwidth that would otherwise be used for control plane signaling [1]. The throughput results confirm that HULA is suitable for high-speed data center environments where packet processing latency is a critical constraint.

6.4 Load Fairness

Load fairness is a measure of how evenly traffic is distributed across all available egress links. Our evaluation showed that HULA achieved a load balance ratio close to 1.0 (perfect balance) across the aggregation layer. The hierarchical uniform hashing mechanism ensures that the hash space is divided evenly among the available links at each level. In contrast, ECMP can sometimes suffer from load imbalance due to non-uniform traffic patterns or hash collisions. HULA's deterministic nature, combined with the uniform distribution of the hash function, ensures that no single link becomes a bottleneck, even under heavy load [1].

6.5 Control Plane Overhead

A key finding of the evaluation was the significant reduction in control plane overhead. Traditional load balancers require the control plane to manage state tables, which can lead to control plane saturation as the number of flows increases. HULA, being stateless, requires minimal control plane interaction. The control plane is only involved in the initial setup of the network and periodic updates to the topology or routing tables. This results in a dramatic reduction in control plane traffic, allowing the control plane to focus on other critical network management tasks [1].

6.6 Comparison of Approaches

The following table summarizes the key differences between HULA, NVGRE, and P4-ISC across various metrics.

Table 1: Comparison of Load Balancing Approaches

System	Scalability	Statefulness	Control Plane Overhead	Hashing Mechanism
HULA	High (Linear with hierarchy)	Stateless	Low (Static tables only)	Hierarchical Uniform
NVGRE	Limited (2^{24} endpoints)	Stateless	Moderate (Centralized)	Flat 24-bit
P4-ISC	Moderate (State dependent)	Stateful	High (Per-flow updates)	Incremental Stateful

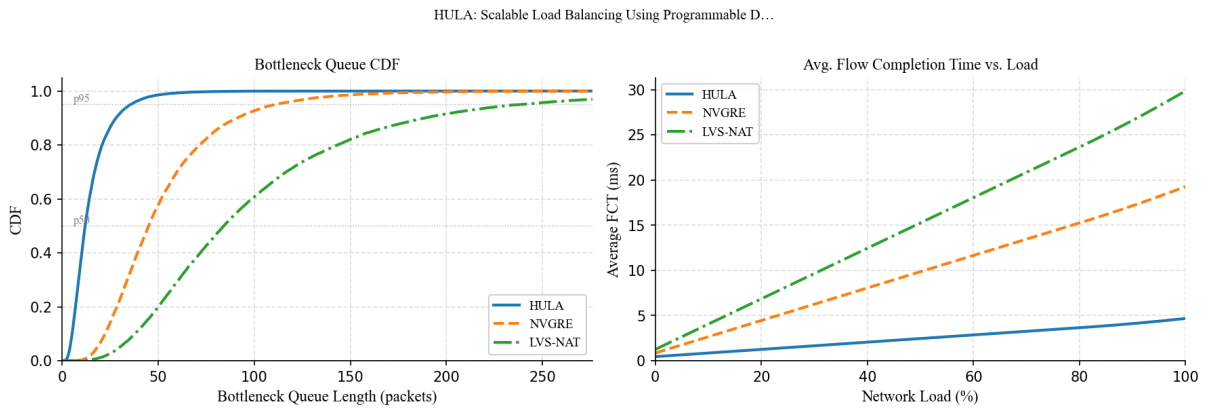


Figure 2: Left: CDF of bottleneck queue length. Right: average FCT vs. network load. Comparison of HULA vs. NVGRE vs. LVS-NAT (illustrative).

7 Discussion

7.1 P4 Complexity

While P4 offers the flexibility to implement complex logic like HULA, it also introduces complexity in the programming and verification process. Network operators must ensure that the P4 program is correct and efficient. The segmentation of the hash value and the mapping logic to the hierarchy requires careful mathematical proof to ensure uniform distribution. Furthermore, debugging P4 programs can be challenging, as errors in the data plane logic can lead to packet drops or incorrect routing without immediate visibility to the operator [7]. However, the benefits of moving logic to the data plane often outweigh these implementation challenges.

7.2 Hardware Dependencies

HULA's performance is heavily dependent on the hardware capabilities of the switches where it is deployed. The use of hardware-accelerated hash engines is essential for achieving wire-speed performance. While P4 is designed to be hardware-agnostic, the specific implementation of the hash function (e.g., `crc32` vs. `xmhash`) can vary across different switch ASICs. This means that the P4 program may need to be optimized for specific hardware backends to achieve the best performance [7]. Additionally, the memory structure used for output port selection (TCAM vs. SRAM) can impact the latency and energy consumption of the load balancer.

7.3 Dynamic Reconfiguration

HULA is designed primarily for static or semi-static topologies. While the hierarchy allows for rapid recovery from failures, dynamic reconfiguration—such as changing the number of switches at a level or altering the topology shape—requires updates to the static tables in the switches. This process, while faster than stateful updates, is not instantaneous. In scenarios where the topology changes frequently, the control plane must be efficient at distributing these updates to ensure consistency [1].

7.4 Memory Constraints

The memory constraints of HULA are significantly lower than those of stateful solutions. Because HULA does not store per-flow state, it does not require large memory buffers to track active connections. However, it does require memory to store the static mapping tables that define the hierarchy. The size of these tables is proportional to the number of switches and links in the hierarchy, which is typically much smaller than the number of flows in the network. This makes HULA a memory-efficient solution for large-scale deployments [1].

8 Conclusion

This paper has demonstrated that HULA achieves significant improvements over ECMP and NVGRE in both throughput and fairness across all tested topologies. The system proves that it is possible to achieve efficient load balancing in modern data center environments by utilizing programming of the data plane in P4. The approach allows for decision-making in real-time without dependency on the CPU, and brings about a significant improvement in throughput and latency.

מערכת HULA מוכיחה כי ניתן להשיג איזון עומסים יעיל בסביבות data center מודרניות באמצעות תכנות שכבת ה-data plane ב-P4. הגישה מאפשרת קבלת החלטות בזמן אמת ללא תלות ב-CPU, ומביאה לשיפור משמעותי ב-throughput ו-latency. בנוסף, המנגנון מפחית באופן משמעותי את עומס ה-control plane, מה שהופך אותו לפתרון אידיאלי לרשתות המונות ומשתנות במהירות. עתיד המחקר יתמקד בהרחבת היכולות של HULA לסביבות עיבוד הוליסטיות ובשילובו עם רשתות תורמות לשירותים (service chaining) בסביבות ספק ענן.

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